

InverPINN

Source-strength bias in physics-informed pollution inversion

Khadis Aidyn / Controlled synthetic research

Can sparse concentration sensors recover both the location and strength of a pollution source?

METHOD AND CONTROLLED DESIGN

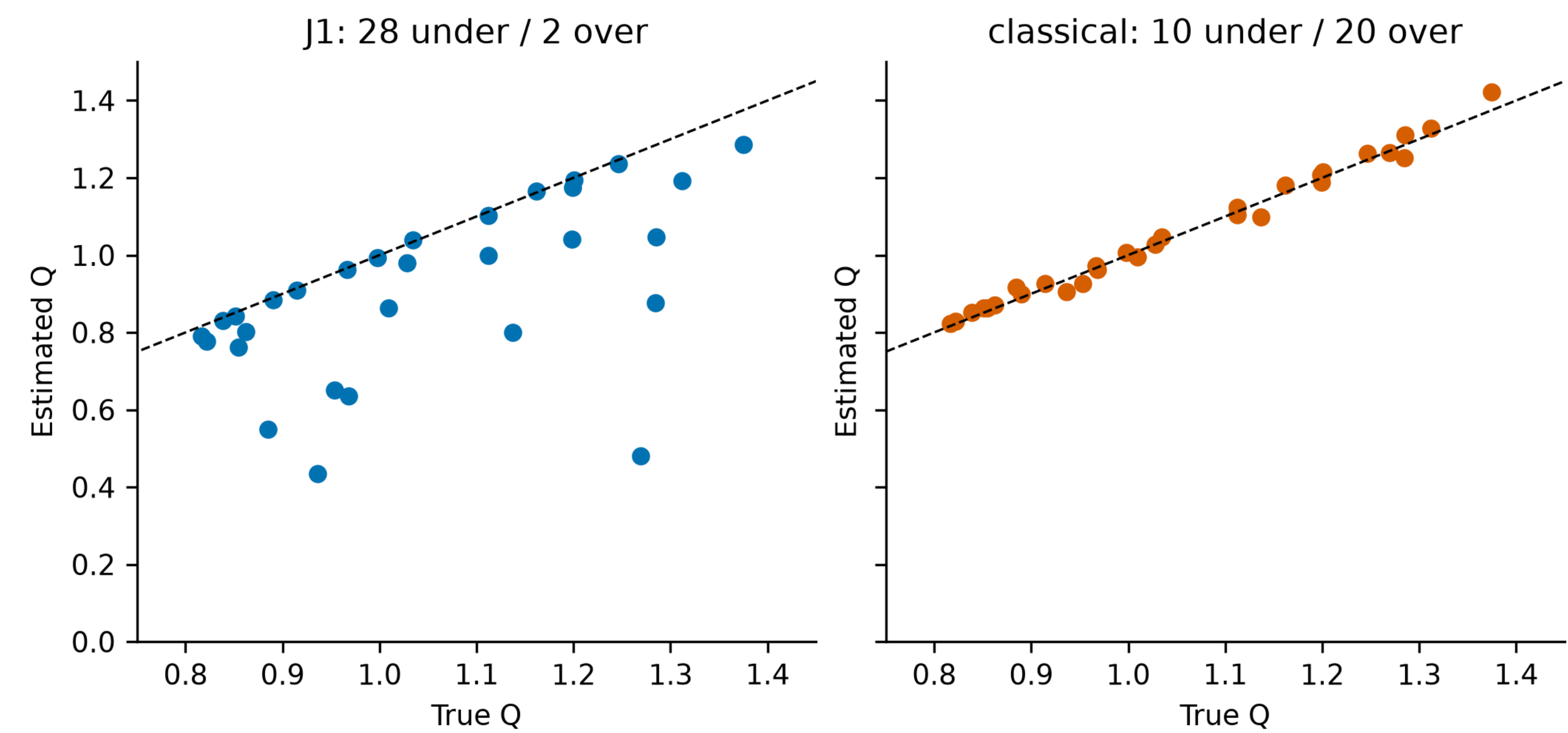
A neural concentration field fits sparse observations and the transport equation.
J1 analytically profiles source amplitude Q from the PDE residual.

$$C_t + uC_x + vC_y = D(C_{xx} + C_{yy}) + QG$$

One Gaussian ($\sigma=0.08$), known wind (0.35, -0.15), $D=0.005$. Unit-square domain. Zero initial and boundary concentration.

20 fixed sensors · 81 times · zero added noise
641×641 reference · 30 untouched final scenarios

SOURCE-STRENGTH DIAGNOSIS



J1 underestimated Q in 28/30 cases (2 overestimates). The identity line shows exact recovery. Q is peak intensity, not integrated emissions.

FINAL BLIND BENCHMARK

J1: 23/30

76.67% recovered; required 27/30 (90%)

FAILED preregistered confirmation

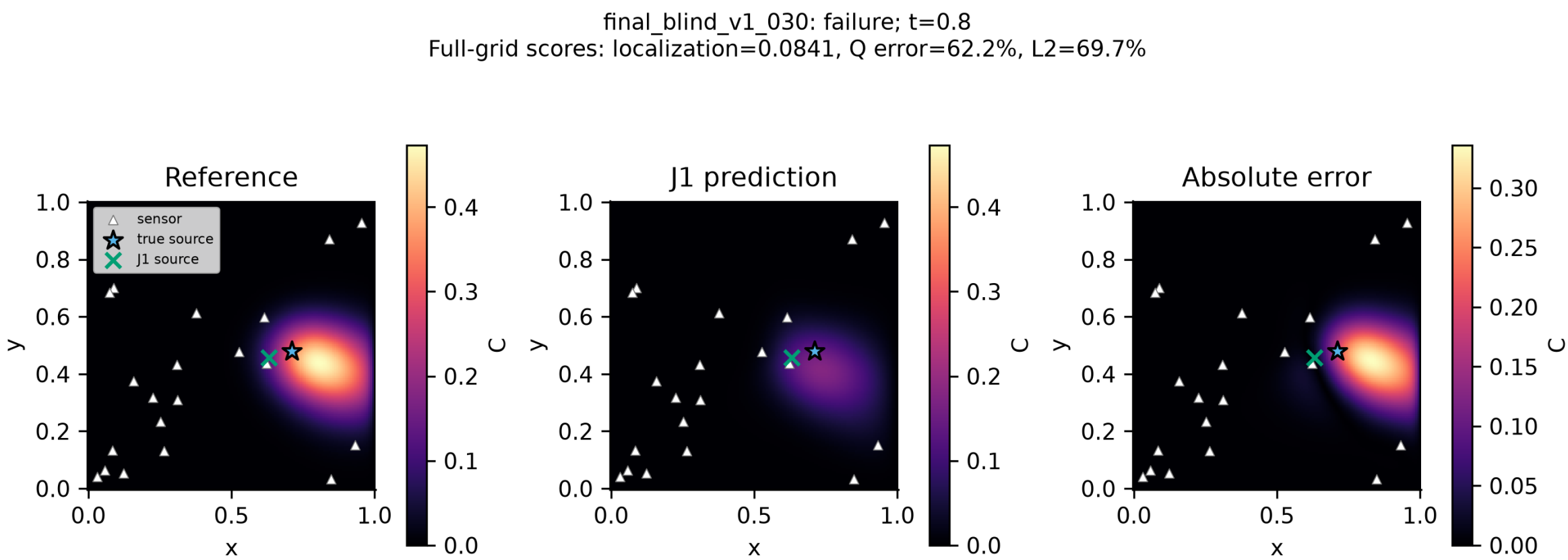
Classical inverse: 30/30

Median error	J1	Classical
Localization	0.005574	0.001111
Relative Q	5.975%	1.218%
Concentration L2	7.423%	1.507%

PHYSICAL VALIDITY DID NOT ENSURE RECOVERY

J1 retained zero negativity and exact IC/BC. All seven joint failures missed the Q criterion despite identical observations for both methods.

WORST LOCALIZATION CASE (030)



Mechanically selected and retained. Source-strength error: 62.183%. Concentration L2: 69.724%.

LIMITS Known transport, one source and zero noise. No validated real Almaty emitter attribution, noisy-sensor robustness or general PINN superiority.